Case Studies in Construction Materials 21 (2024) e03991

Contents lists available at ScienceDirect

Case Studies in Construction Materials

journal homepage: www.elsevier.com/locate/cscm

Study on the effect of recycled polyvinyl chloride powder on the properties of cementitious composite

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ARTICLE INFO

Keywords:
Cementitious composite
Recycled polyvinyl chloride (PVC) aggregate
Fine aggregate
Mechanical property
Durability

ABSTRACT

A novel type of cementitious composite was proposed and studied, where recycled polyvinyl chloride (PVC) aggregate was employed as fine aggregate to partially replace the fine aggregates of sand in varying proportions. The fine aggregates of sand were replaced at five volume ratios (5 %, 10 %, 20 %, 30 %, and 50 %). A series of experiments were carried out to evaluate the workablity, mechanical properties and durability of the proposed cementitous coposites with PVC. The scanning electron microscopy (SEM) was applied to study the relationships between the mechanical properties and microstructures. The results showed that the fluidity increased by 12.5 % with a 50 % PVC replacement. However, the dry density decreased by 5 %, and water absorption increased with a higher PVC content. The compressive strength declined by 3.7 %, 6.1 %, 6.3 %, 11.5 %, and 28.8 % as the replacement ratio increased from 5 % to 50 %; while, the flexural strength decreased by 3.7 %, 6.8 %, 9.8 %, 17.0 %, and 23.5 %, respectively. The relative dynamic elastic modulus showed a 12.2 % reduction at 50 % PVC replacement. These decreases are attributed to the weaker bonding between PVC particles and the cement matrix, increased porosity, and micro-cracks at the interface transition zone, as confirmed by the SEM examination. The microstructural analysis revealed more porous and cracked interface zones with a higher PVC content, which impacted the mechanical properties of the composites. Despite the reduction in the mechanical strength, the durability of the composite improved significantly, with an enhanced resistance to chloride ion penetration and freeze-thaw cycles.

1. Introduction

Common plastics are high molecular weight polymers formed by adding chemical additives under certain conditions. They offer advantages such as stable chemical properties, good plasticity, and ease of processing [1]. Plastics play an essential role in various industries, including packaging, construction, and healthcare, but their widespread use has resulted in substantial quantities of plastic waste [2–4]. Plastics are difficult to degrade naturally, leading to environmental challenges, including pollution and the accumulation of waste in oceans and landfills. This situation underscores the urgent need to find sustainable solutions for managing plastic waste. One promising approach is recycling plastics into construction materials, such as cementitious composites, which not only address

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https://doi.org/10.1016/j.cscm.2024.e03991
Received 7 March 2024; Received in revised form 31 October 2024; Accepted 14 November 2024
Available online 15 November 2024
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waste management but also reduce the consumption of natural resources like sand and gravel, essential for construction [5–7].

Although there is growing research on incorporating recycled plastics into cementitious composites, most studies have focused on the use of polyethylene terephthalate (PET) and polypropylene (PP) as fibers or aggregates [8–15]. Research has predominantly focused on the fundamental mechanical properties of recycled plastic concrete, emphasizing the type, shape, particle size, dosage, and pretreatment methods of waste plastic aggregates. Additionally, some scholars have examined the construction technology of recycled plastic concrete and its performance in practical applications [16–20]. For instance, Kou, Lee, Poon, and Lai [21] developed lightweight concrete using waste PVC particles and expansive clay, achieving low density, high ductility, and good resistance to chloride ion erosion. Similarly, Kim, Yi, Kim, Kim, and Song [22] and Ochi, Okubo, and Fukui [23] incorporated waste polyethylene terephthalate (PET) plastics as short fibers into structural concrete. Their tests indicated that while the compressive strength and elastic modulus decreased with increased PET fiber volume fraction, the shrinkage and ductility of the composites improved significantly. Reis and Carneiro [24] used PET aggregate to replace fine aggregate in cementitious composites, finding that although the compressive and flexural strengths decreased, the elastic modulus and toughness improved with PET content. Fakhri, Shahryari, and Ahmadi [25] studied the mechanical properties of recycled PVC aggregate in asphalt mixtures, discovering that it enhanced the softening point, viscosity, and specific gravity of asphalt while reducing penetration and ductility. This led to improved rutting resistance and creep stiffness at high temperatures, showcasing good road performance.These studies primarily investigated the mechanical properties and construction technology of recycled plastic concrete. However, research on polyvinyl chloride (PVC), one of the most widely used plastics globally, remains limited. PVC has distinct chemical properties, including high resistance to chemical attacks and a high molecular structure, making it an intriguing candidate for recycling in the construction industry. Despite these benefits, the integration of recycled PVC as a fine aggregate replacement in cementitious composites has not been extensively studied [13–15]. Thus, this study aims to fill this research gap by exploring the effects of using recycled PVC as a fine aggregate on the mechanical properties, durability, and serviceability of cementitious composites.

The selection of recycled PVC as a fine aggregate replacement is based on several key factors. PVC is abundantly available due to its widespread use in various industries, including construction, healthcare, and packaging. The recycling of PVC helps to alleviate the environmental burden by reducing the volume of plastic waste that would otherwise end up in landfills or incineration, both of which have harmful environmental impacts [16,18]. Moreover, from an economic perspective, recycled PVC is a cost-effective alternative to traditional fine aggregates such as natural sand, which is becoming increasingly scarce [26–28]. Furthermore, using recycled PVC can contribute to the development of lightweight cementitious composites due to PVC's lower density compared to conventional aggregates, which may have advantages in structural applications where weight reduction is desirable.

Durability and serviceability are critical properties for cementitious composites, especially when incorporating recycled materials like PVC. Durability refers to the ability of the material to withstand environmental degradation, such as chloride ion penetration and freeze-thaw cycles, which are key factors in ensuring long-term performance. Durability is particularly important in recycled PVC composites because of the potential for plastic-based aggregates to weaken the cement matrix over time. However, PVC's resistance to chemical degradation may offer advantages in harsh environments, such as resistance to chloride ion penetration and freeze-thaw cycles, which are critical for long-term performance in infrastructure projects. Serviceability, on the other hand, relates to the material's ability to maintain mechanical properties and resist deformation over time. These properties are particularly important when using recycled PVC because plastic-based aggregates can potentially weaken the mechanical strength of the composite. Therefore, understanding the trade-offs between durability and mechanical performance is essential for determining the feasibility of using recycled PVC in sustainable construction applications.

The primary objectives of this study are threefold: (1) to assess the influence of varying levels of recycled PVC aggregate on the physical and mechanical properties of cementitious composites, including fluidity, dry density, water absorption, compressive strength, flexural strength, and relative dynamic elastic modulus; (2) to evaluate the durability of the composites, particularly their resistance to chloride ion penetration and freeze-thaw cycles; and (3) to conduct a microstructural analysis using scanning electron microscopy (SEM) to understand the internal behavior of the composites. This comprehensive evaluation will provide insight into the performance of recycled PVC-based cementitious composites and their potential contribution to sustainable construction practices.

To ensure the accuracy and comparability of the results, this study followed relevant standards and guidelines. The compressive and flexural strengths were measured according to Chinese standard GB/T17671–2021, while the chloride ion penetration resistance was assessed using the colorimetric method outlined in JTS/T 236–2019. Freeze-thaw resistance tests were conducted following the procedures specified in GB/T50082–2009.

Table 1
The chemical compositions of the cement.

Materials	SiO_2	Al_2O_3	MgO	CaO	Fe_2O_3	SO_3	Loss on ignition
Chemical composition (%)	21.80	5.25	2.33	60.76	2.70	2.12	2.76

2. Experimental design

2.1. Materials

2.1.1. Cement

Cement was the primary cementing material in cementitious composite preparation. The type and strength of cement significantly influence the properties of concrete. P·O 42.5 ordinary Portland cement from Xiangtan Jiancheng Company was used in this experiment. The chemical composition of the cement is presented in Table 1, and its physical properties, including specific surface area, setting time, and loss on ignition, are outlined in Table 2. These detailed characterizations ensure consistency in the cement's performance.

2.1.2. Fly ash

Fly ash was used to replace part of the cement in the concrete, which can reduce the amount of cement, improve the workability of the concrete mixture, and increase the permeability of the concrete. Fly ash was grade I, sourced from Henan Platinum Refractory Company. Its chemical composition, including SiO_2, Al_2O_3, and other oxides, is detailed in Table 3, while its physical properties, such as bulk density and fineness, are shown in Table 4.

2.1.3. Silica fume

Silica fume, used as a supplementary material, was produced by Hunan Solid Engineering New Materials Company. The chemical composition and particle size distribution of the silica fume are provided in Tables 5 and 6, respectively.

2.1.4. Fine aggregate

The fine aggregate used in the test is white quartz sand with a particle size of 20–40 mesh, and the main component is silicon dioxide.

2.1.5. Recycled PVC aggregate

The recycled PVC aggregates used in this study were provided by Dongyan Huachuang Polymer Technology Company and were white with a particle size of 20 mesh (0.84 mm). No additional pre-treatment was applied to the PVC aggregates before incorporation into the composite. The rationale for using 20-mesh PVC was based on the size compatibility with the fine aggregate traditionally used in cementitious composites, ensuring proper bonding and distribution in the mix. The recycled PVC aggregate is shown in Fig. 1.

2.1.6. Water reducer

A polycarboxylate superplasticizer was used as a water reducer to obtain suitable workability for the cementitious composite. Its water reduction rate was 20 %. According to the pre-tests, the suitable dosages of water reducer were between 0.1 % and 0.5 %.

2.1.7. Water

The applied water was pure in the laboratory.

2.2. Sample preparation

The R-PVC aggregate was mixed in the experiments by replacing part of the fine aggregate with an equal volume. The fine aggregate used was sand with a density of 2650 kg/m^3, while the density of the recycled PVC aggregate was 1600 kg/m^3. The control group, labeled NM, was prepared with a fixed water-cement ratio of 0.25. The fine aggregate in the cementitious composite was replaced by recycled PVC aggregates at replacement ratios of 5 %, 10 %, 20 %, 30 %, and 50 %. These specific replacement percentages were selected to evaluate the material's behavior at low, moderate, and high levels of PVC incorporation, allowing us to observe a broad spectrum of mechanical and durability performance variations. Additionally, previous studies on recycled plastics suggest that these ratios cover a range that captures both minor and significant changes in the composite's properties.

The control group, labeled NM (Normal Mortar), consisted of a standard cement mortar mixture without any PVC replacement. The composition of the control mix is presented in Table 7, which includes the cement, fly ash, silica fume, and fine aggregate proportions. The purpose of the control group is to provide a baseline against which the mechanical, physical, and durability properties of the recycled PVC composites can be compared.

Table 2
Cement performance indicators.

Strength grade	Specific surface area (m^2/kg)	Initial setting time (min)	Final setting time (min)	Loss on ignition (%)
P.O 425	331	147	224	4.27

Table 3
The chemical compositions of the fly ash.

Materials	SiO_2	Al_2O_3	MgO	Fe_2O_3	CaO	Na_2O	TiO_2	Loss on ignition
Chemical composition (%)	53.36	29.81	0.61	4.57	5.47	0.37	1.84	2.47

Table 4
Physical performance indicators of of the fly ash.

Index	Test results	International requirements
Apparent density (kg/m^3)	2437	/
Bulk density (kg/m^3)	920	/
Fineness (0.0045 mm square sieve hole)	18	≤25.0
Loss on ignition(%)	2.49	≤8.0

Table 5
Physical performance indicators of of the fly ash.

Materials	SiO_2	MgO	CaO	K_2O	Fe_2O_3	Na_2O	Al_2O_3	Loss on ignition
Chemical composition (%)	92.61	0.29	1.86	0.98	0.57	0.17	0.85	2.24

Table 6
Physical performance indicators of of the fly ash.

Particle size/μm	< 2	< 4	< 10	< 16	< 20	< 32	< 64	< 80	< 100
Volume content (%)	98.23	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0



Fig. 1. Recycled PVC aggregates.

Table 7
Mix the proportion of recycled PVC mortar.

Number	Cement	Fly ash	Silica fume	Fine Aggregate	water	Water reducer	Recycled PVC aggregate
NM	691	502	62	628	314	2.95	0
PM−20–5	691	502	62	596.6	314	2.95	16.4
PM−20–10	691	502	62	565.2	314	2.95	32.7
PM−20–20	691	502	62	502.4	314	2.95	65.4
PM−20–30	691	502	62	439.6	314	2.95	98.0
PM−20–50	691	502	62	313.2	314	2.95	163.5

2.3. Experimental variability and error control

To manage experimental variability and minimize potential sources of error, all tests were conducted under controlled laboratory conditions. Temperature and humidity were kept consistent throughout the curing and testing phases. Each test was performed in triplicate, and the average value was reported. In cases where the standard deviation exceeded 5 % of the average, additional tests were performed to ensure accuracy.

Variability in the PVC aggregate particle size was minimized by using aggregates with uniform 20-mesh size. Additionally, all raw materials were weighed precisely using an electronic balance with a precision of 0.01 g. These steps ensured that variations in the material composition and preparation process did not influence the final results.

2.4. Method

Before testing various indicators, it is necessary to prepare recycled PVC cement mortar. Impurities such as gravel should be removed from the sand, ensuring that all materials used in the experiment, including sand and recycled PVC aggregates, are dry. In this study, a JJ-20H cement mortar mixer was used to prepare the recycled PVC cement mortar, as shown in Fig. 2(a). The specific mixing process is as follows:

1. Accurately weigh the required raw materials according to the experimental mix proportions.
2. Wet the mixing blades and the mixing bowl with water.
3. Sequentially add cement, fly ash, and silica fume into the mortar mixing bowl and dry mix for 60 seconds to ensure uniform blending of the cementitious materials.
4. Mix the water reducer, quartz sand, and recycled PVC aggregate, then gradually add the mixture into the mixing bowl while keeping the mixer running at a low speed. Finally, add water and mix at high speed for 60 seconds.
5. Pause mixing for 30 seconds, scrape any adhered recycled PVC cement mortar from the mixer and the bottom of the mixing bowl into the bowl, and continue mixing at high speed for another 60 seconds.

After mixing, pour the cement mortar into the corresponding molds and use a cement mortar vibrating table to vibrate the specimens for 1–2 minutes. During the vibration process, remove any excess mortar. After vibrating, smooth the surface of the specimens, wrap the molds with plastic wrap, label them, and place them in a standard curing chamber. The molds are typically demolded after 24 hours, and the specimens are then placed in a concrete standard curing chamber to cure until the designated age. The standard curing chamber is shown in Fig. 2(b).

2.4.1. Fluidity

According to the Test Method for Fluidity of Cement Mortar (GB/T2419−2005), the fluidity of the recycled PVC cement mortar was tested using a fluidity cylinder. Given the excellent fluidity of the prepared recycled PVC mortar, which demonstrated outstanding fluidity without the use of a flow table, a mortar fluidity tester was employed instead. Each mix proportion was measured three times, and the average of the three test values was recorded as the fluidity result. If the deviation of any individual test value exceeded 10 mm compared to the other two, that test value was discarded.



(a) JJ-20H cement mortar mixer (b) Ordinary concrete standard curing box

Fig. 2. Preparation of recycled PVC cement mortar specimens.

2.4.2. Dry density

The dry density test was conducted using 40 mm × 40 mm × 40 mm cubic specimens. After curing the specimens for 28 days, they were removed and allowed to air dry naturally. Subsequently, the specimens were placed in an oven set to a temperature of 105 ± 5 °C for drying. The process of weighing and drying was repeated until the weight of the specimen remained constant. The final weight measured at this point was considered the dry mass of the concrete. The calculation formula is as follows:

$$\rho = \frac{\mathrm{m}}{v} \tag{1}$$

Where ρ is the dry density of the specimen (kg/m^3), m is the dry mass of the test block (g), and v is the apparent volume of the block (mm^3).

2.4.3. Water absorption

The water absorption test of recycled PVC cement mortar refers to the Standard for Test Methods of Basic Properties of Building Mortar (JGJ/T70–2009). The main instruments used in the test include a balance, an oven, a water tank, and several plastic rods. The oven used is a 101–34 type electric blast constant temperature drying oven with drying chamber dimensions of 500 mm × 600 mm × 750 mm and a temperature range of 10 ℃ to 300 ℃. The test specimens used are 40 mm × 40 mm × 40 mm in size, with three specimens per group. After standard curing for 28 days, the specimens are removed and placed in the oven set to a temperature of 105 ± 5 ℃, dried to constant weight, and then weighed to record their mass as m_d. After weighing, water is added to the water tank, and plastic rods are placed at the bottom. The specimens are placed in the water tank with the smooth surface facing down and soaked for 24 hours. After soaking, the specimens are removed, wiped dry with a cloth, and their mass is measured and recorded. The water absorption calculation formula is as follows:

$$W_a = \frac{m_s - m_d}{m_d} \times 100\% \tag{2}$$

Where W_a is the water absorption rate of R-PVC-CC (%), m_s is the saturated water absorption mass of mortar (g), and m_d is the mass of mortar after drying (g).

2.4.4. Compressive strength

The compressive strength of the specimens was tested according to Chinese standard (GB/T17671–2021). The specimens were 40 mm × 40 mm × 40 mm in size. After the mortar was poured, it was compacted using a vibrating table and covered with plastic film to prevent cracking during the hardening process. The specimens were demolded after 24 hours and transferred to a standard curing chamber. After reaching the respective curing ages, the compressive strength values of each test group were measured at 3, 7, and 28 days. The tests were conducted using a TYA-300B microcomputer-controlled cement flexural and compressive testing machine, as shown in Fig. 3. The specific experimental steps are as follows:

1. Remove the specimens cured to the respective ages from the standard curing chamber, clean off any debris from the specimen surfaces, and air-dry at room temperature for 2 hours.
2. After cleaning the debris from the upper and lower platens of the pressure testing machine, place the specimen on the platen, ensuring it is centered and that the contact surfaces between the specimen and the machine are level.
3. Start the testing machine, set the test number, corresponding age, and loading rate, and ensure that the loading process is continuous, uniform, and slow.



Fig. 3. Compressive strength measuring equipment.

4. When the specimen undergoes significant deformation, stop the machine and record the load at that moment.

The compressive strength of the cement mortar calculation formula is as follows:

$$f_c = \frac{F}{A}\# \tag{3-1}$$

Where f_c is the compressive strength of cement mortar specimen (MPa), F is the specimen failure load (N), and m_d is the actual compressive area of specimen (mm^2).

2.4.5. Flexural strength

The flexural experiments were conducted according to Chinese standard (GB/ T17671–2021). The specimen size was 40 mm × 40 mm × 160 mm. The flexural strength values at curing periods of 3d, 7d, and 28d were measured. The TYA-300 B automatic cement flexural and compressive strength testing machine was employed to test the flexural strength. The specific experimental steps are as follows:

1. Remove the specimens cured to the respective ages and allow them to air dry.
2. Ensure that the casting surface of the specimen is not used as the bearing surface, and place the specimen in the center of the bottom support, maintaining it level.
3. Start the testing machine, adjust the specimen parameters, and set the loading rate to 50±10 N/s, applying the load uniformly until the specimen fails.
4. After the test is completed, stop the testing machine and record the load value at the moment of failure.

The flexural strength of the cement mortar calculation formula is as follows:

$$R_f = \frac{1.5F_fL}{b^3}\# \tag{3-2}$$

Where R_f is the flexural strength of cement mortar (MPa), F_f is the load value when the specimen breaks (N), L is the distance between the cylinders of the testing machine support (mm), and B is the side length of the square section of the prism (mm).

2.4.6. Relative dynamic elastic modulus

The relative dynamic elastic modulus was measured according to Chinese standard (GB/T50082–2009). The size of the specimen was 40 mm × 40 mm × 160 mm, and the DT-20 relative dynamic elastic modulus tester was used for testing, as shown in Fig. 4.

The specific experimental steps are as follows:

1. Remove the specimens cured for 28 days from the standard curing chamber, allow them to air dry in the laboratory, and weigh the specimens.
2. Adjust the relevant parameters of the DT-20 dynamic modulus tester. Apply petroleum jelly to both ends of the specimen to ensure better contact between the receiver and transmitter probes and the specimen, resulting in more accurate measurements.
3. Secure the transmitter and receiver probes at the same positions on both ends of the specimen and begin the test.



(a) DT-20 relative dynamic elastic modulus tester (b) Test process

Fig. 4. Relative dynamic elastic modulus test.

2.4.7. Chloride ion erosion

The chloride ion erosion test was measured using the color development method specified in Chinese standard (JTS/T 236–2019). The specimen size was 70.7 mm×70.7 mm×70.7 mm. After curing the specimens for 28 days, they were removed and air-dried before beginning the experiment. The chemical reagents used in the experiment are shown in Fig. 5(a). The specific steps are as follows:

1. Prepare the NH4Cl solution with a concentration of 6 mol/L. Place cubic plastic rods at the bottom of the water tank, position the specimens properly, and pour the solution until it covers the specimens by approximately 10 mm. Seal the water tank with plastic wrap, as shown in Fig. 5(b).
2. Prepare the test solutions for the colorimetric method, including 0.1 mol/L potassium chromate solution, 1 % ethanol phenolphthalein solution, and 0.1 mol/L silver nitrate solution. After curing the specimens to the respective ages, remove them and send them to the Xiangtan City Building Materials Market for cutting along the midline into two halves. Air-dry the cut specimens before testing.
3. For each group of 6 specimens, place 3 specimens in an oven at 105 ℃ for 0.5 hours, then cool them for 1 hour. Spray the cut surfaces with potassium chromate solution. Repeat this process twice more, allowing the specimens to air-dry naturally after the third time, and then immediately spray with silver nitrate solution. For the other 3 specimens, air-dry naturally and spray with ethanol phenolphthalein solution.
4. Once the specimens are air-dried, measure the erosion depth using an electronic caliper. Measure 4 points on the cut surface of each specimen, totaling 16 points for each group, and take the average erosion depth as the representative value for the group. At this point, the cut surface sprayed with silver nitrate solution will appear dark red, while the surface sprayed with ethanol



Fig. 5. Colorimetric test.

phenolphthalein solution will show pink, with the surrounding gray-white areas indicating the erosion depth, as shown in Fig. 5(c) and (d).

2.4.8. Freeze-thaw resistance

The freeze-thaw resistance experiments were conducted according to Chinese standard (GB/T50082–2009). The freeze-thaw medium in the experiments was tap water in the laboratory. Considering the loss of raw materials and the size of the freeze-thaw machine studio, the size of the R-PVC-CC specimen in the frost resistance experiments was 40 mm × 40 mm × 160 mm. The instrument was a TDR-3 concrete rapid freeze-thaw test machine. As shown in Fig. 6, the evaluation indices for the freeze-thaw resistance of recycled PVC cement mortar are the relative dynamic elastic modulus and the mass loss rate of the specimens. The main steps are as follows:

1. Immerse the specimens cured for 28 days in water for 24 hours, with plastic rods placed under the specimens.
2. Before placing the specimens into the freeze-thaw testing machine, take photos and measure the mass of each specimen before freeze-thaw cycling (M_0), then use a dynamic modulus tester to measure the initial relative dynamic elastic modulus (G_0) of each specimen.
3. Set the parameters of the rapid freeze-thaw machine: the freezing temperature is set to −20 ± 5 ℃ for 4 hours, and the thawing temperature is set to 20 ± 5 ℃ for 4 hours. One freeze-thaw cycle is 8 hours.
4. Measure the mass (Mi) and the relative dynamic elastic modulus (Gi) of the specimens after every 25 freeze-thaw cycles.
5. The freeze-thaw cycling test is terminated when any of the following conditions are met:

a. 100 freeze-thaw cycles are completed;
b. The relative dynamic elastic modulus of the specimen decreases to 60 %;
c. The mass loss rate of the specimen reaches 5 %.

After the freeze-thaw cycling test, calculate the loss of relative dynamic elastic modulus according to Eqs. 4–1, accurate to 0.01 MPa.

$$\Delta G_c = \frac{G_0 - G_i}{G_0} \times 100\%\# \tag{4-1}$$

Where ΔG_c is relative dynamic elastic modulus loss rate of recycled PVC cement mortar after i freeze-thaw cycles (%), G_0 is initial relative dynamic elastic modulus of recycled PVC cement mortar specimen before freeze-thaw cycles (N), G_i is the relative dynamic elastic modulus of recycled PVC cement mortar specimen after i freeze-thaw cycles (MPa).

The mass loss rate is calculated according to formula 4–2:

$$\Delta M_c = \frac{M_0 - M_i}{M_0} \times 100\%\# \tag{4-2}$$



(a) TDR-3 concrete rapid freeze-thaw test machine (b) Specimen and specimen box

Fig. 6. Freeze-thaw cycle test of recycled PVC cement mortar.

Where ΔM_c is the mass loss rate of recycled PVC cement mortar after i freeze-thaw cycles (%), M_0 is mass of recycled PVC cement mortar specimen before freeze-thaw cycles (N), M_i is the relative dynamic elastic modulus of recycled PVC cement mortar after i freeze-thaw cycles (MPa).

2.4.9. Scanning electron microscope

The microstructure was measured with scanning electron microscopy (SEM). The internal characteristics of the recycled PVC aggregates were observed using a Caisi Sigma300 electronic scanning microscope. As shown in Fig. 7, the specimens were soaked in anhydrous ethanol and then dried. The surfaces were then sputter-coated with platinum to ensure conductivity. The specimens were placed on the sample stage for observation. The primary focus was on the interface between the recycled PVC aggregate and the matrix transition zone, internal cracks within the matrix, and the characteristics of the internal pores.

The preparation of specimens and experimental details are shown in Fig. 8.

3. Results and discussion

3.1. Fluidity

In practical construction, mortar should possess excellent fluidity, with good adhesion between components and no bleeding, to facilitate pumping and casting. Azad et al. [29] showed that incorporating waste PVC aggregates can effectively improve the fluidity of PVC cement mortar, provided the incorporation amount does not exceed 30 %. In contrast, Manjunatha et al. [30] reported that when recycled PVC replaces the cementitious materials in the mortar, the fluidity of the cement-based material decreases, with a significant reduction in mechanical properties at low incorporation levels. Currently, there are conflicting results regarding the effect of recycled PVC incorporation on the fluidity of cementitious materials. The primary reason for these discrepancies is the different sources and particle sizes of the recycled PVC aggregates used in various studies. Bartolomeo et al. [31] conducted XRD analysis on waste plastic cement mortar and found that the incorporated waste plastic aggregates do not participate in the hydration reaction of cement-based materials and therefore do not form new substances. The incorporation of waste plastic aggregates primarily affects the formation of the recycled PVC-cement paste interface transition zone, thereby altering the properties of the cement-based materials. Therefore, this study investigates the effects of different recycled PVC incorporation levels.

The results of the R-PVC-CC fluidity are shown in Fig. 9. The addition of recycled PVC aggregates significantly increases the fluidity of the cementitious composite. The fluidity of the PM-20–50 mix increased by 12.5 % compared to the control group. This enhancement in fluidity is primarily due to the hydrophobic nature of PVC, which does not absorb water, leaving more free water in the mix. Similar findings were reported by Azad et al. [29], who observed an increase in fluidity with plastic aggregate incorporation, though their study focused on PET. In contrast, Manjunatha et al. [30] found a reduction in fluidity when recycled plastic replaced a portion of the cementitious materials. This highlights the difference in behavior between various types of recycled plastics.

3.2. Dry density

The dry density of concrete refers to the ratio of the concrete mass to its apparent volume in a dry state. Studies have shown that the dry density of concrete has a significant influence on the concrete's water absorption, softening coefficient, compressive strength, thermal conductivity and other properties.

The dry density results of the R-PVC-CC are shown in Fig. 10. The density gradually decreased with the increase in the PVC replacement rate. The dry density of the R-PVC-CC decreased by 5 % at a 50 % PVC replacement ratio. This reduction is expected due to the lower density of PVC (1600 kg/m^3) compared to quartz sand (2650 kg/m^3). This trend is consistent with Fakhri et al. [25], who



Fig. 7. Scanning electron microscope.

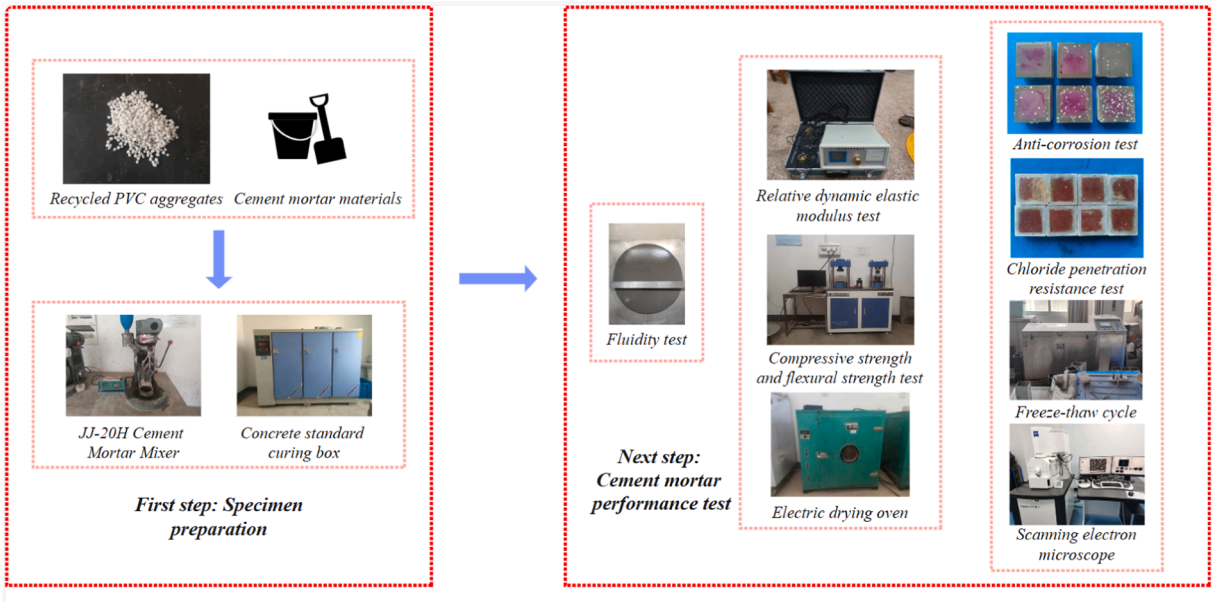


Fig. 8. Experimental details.

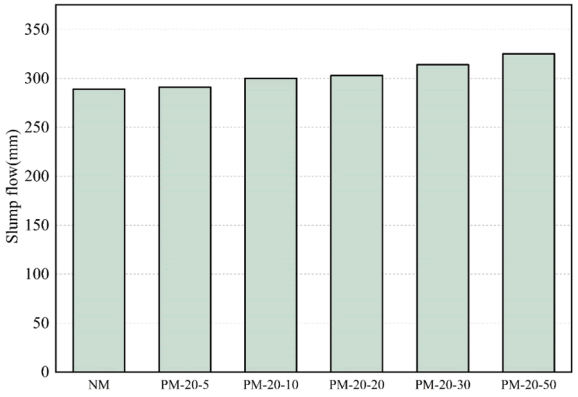


Fig. 9. Fluidity.

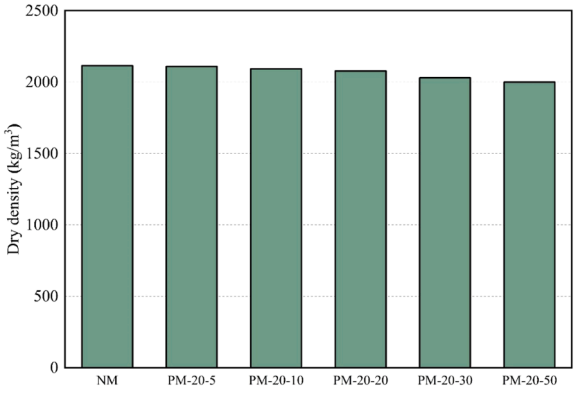


Fig. 10. Dry density test results.

also observed a decrease in density when using recycled PVC in asphalt mixtures. This substantial difference in density means that when recycled PVC aggregates replace quartz sand in the mixture, the overall mass of the composite decreases while the volume remains constant, leading to a reduction in dry density.

3.3. Water absorption

Fig. 11 shows the results of water absorption of R-PVC-CC. Water absorption increased with the increase in recycled PVC aggregate content, with a 15 % increase at a 50 % replacement ratio. This increase is attributed to the higher porosity of the cement matrix caused by the poor bonding between the hydrophobic PVC aggregates and the cement paste. As observed in similar studies, such as Reis and Carneiro [24], the inclusion of recycled plastics generally increases porosity, leading to higher water absorption. The microstructural analysis further supports this, showing more significant voids at the interface between the PVC and the cement paste.

3.4. Compressive strength

The compressive strengths of R-PVC-CC are shown in Fig. 12. The compressive strength decreased significantly with the increased recycled PVC aggregate replacement rate. The compressive strength at 28 days decreased by 3.7 %, 6.1 %, 6.3 %, 11.5 %, and 28.8 %, respectively. Notably, the effect of recycled PVC aggregate on the early strength of the cementitious composite is slightly more substantial than that on the 28-day strength.

The observed decrease in compressive strength with increasing recycled PVC aggregate content can be attributed to several factors. Firstly, PVC particles, as a plastic material, have lower strength compared to the quartz sand in cement mortar and do not possess effective load-bearing capacity. Therefore, incorporating PVC particles effectively reduces the volume of load-bearing aggregates in the cement mortar, thus decreasing the load-bearing area of the cement matrix. Secondly, due to the hydrophobic nature of PVC particles, their bonding effect with cement mortar is inferior to that of quartz sand. This results in the recycled PVC-cement stone interface becoming a defect area in the transition zone, which has poor load-bearing capacity, thereby reducing the strength of the recycled PVC cement mortar. Thirdly, the material difference between recycled PVC particles and cement mortar, coupled with the hydrophobicity of PVC particles, leads to the presence of numerous micro-bubbles in the recycled PVC-cement stone interface. These bubbles, along with the distribution of recycled PVC particles within the cement mortar, create a higher porosity in the recycled PVC cement mortar. It is well known that the more porous the cement matrix, the weaker its load-bearing capacity. Therefore, the increase in porosity is also a significant factor contributing to the reduction in the strength of the cement matrix.

These findings align with previous research that has reported similar trends in the mechanical properties of cementitious composites incorporating recycled materials. For instance, studies by Reis J M L [32] and Kim S B [33] have also observed a reduction in compressive strength with the inclusion of recycled aggregates, attributing it to poor aggregate-matrix bonding and lower aggregate strength, which supports our findings regarding the diminished bonding effect and increased defect areas. However, Compressive strength decreased by 28.8 % at 50 % PVC replacement, which is significantly higher than the 15–20 % reductions reported by Kou et al. [21]. The larger reduction in this study can be attributed to the lack of pre-treatment of the PVC aggregates. PVC aggregates do not actively participate in the hydration process and act as inert fillers, leading to a weaker interface between the aggregates and the cement matrix. This weak interface, as seen in the SEM analysis, is prone to crack initiation, further reducing compressive strength.

3.5. Flexural strength

The effect of recycled PVC aggregate replacement rate on the flexural strength of the cementitious composite is shown in Fig. 13. The impact of recycled PVC aggregate on the flexural strength of the cementitious composite is similar to its effect on compressive strength. Specifically, the 28-day flexural strength decreased by 3.7 %, 6.8 %, 9.8 %, 17.0 %, and 23.5 %, respectively, as the recycled

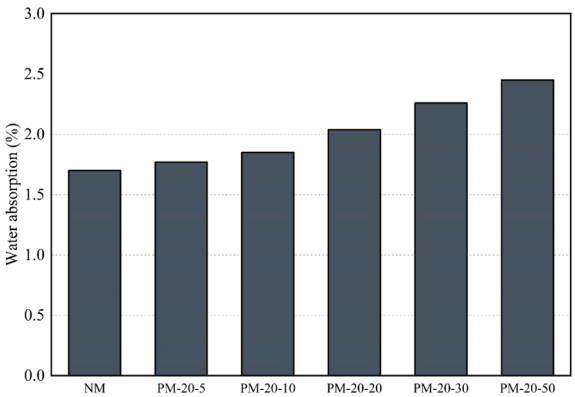


Fig. 11. Water absorption test results.

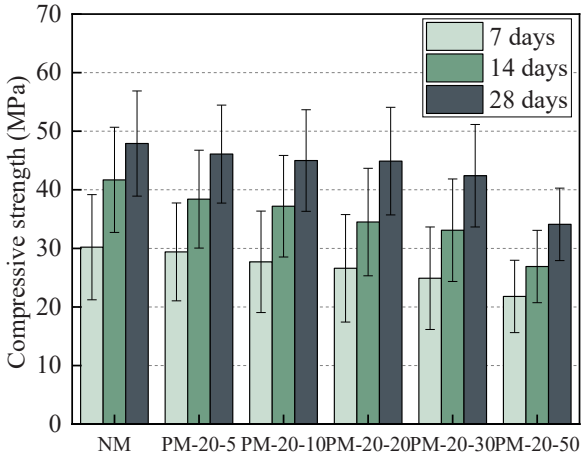


Fig. 12. Results of compressive strength of the specimens.

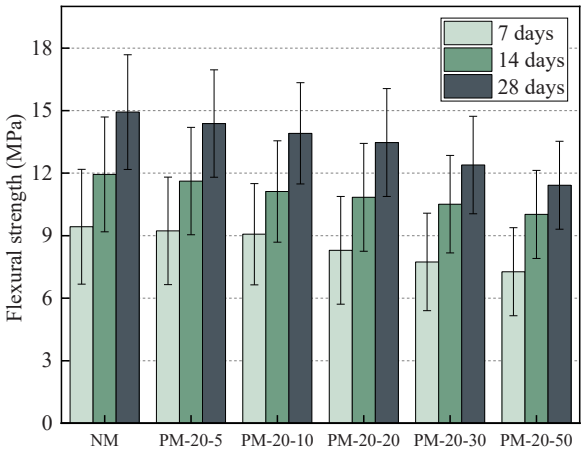


Fig. 13. Results of the flexural strengths of the specimens.

PVC aggregate replacement rate increased.

Flexural strength decreased by 23.5 % at 50 % PVC replacement, which is comparable to findings by Kim et al. [22] where PET fibers led to similar reductions. Unlike PET fibers, which contribute to crack bridging and improved ductility, PVC in granular form

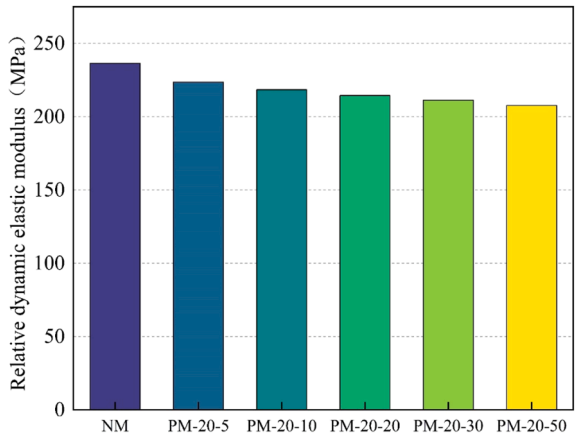


Fig. 14. Relative dynamic modulus of elasticity of the specimens.

does not offer such benefits, resulting in a more brittle material. This explains the more significant reduction in flexural strength observed in this study compared to those using fibrous plastic materials.

The effect of recycled PVC aggregate on flexural strength is less pronounced than its effect on compressive strength. For the 3-day and 7-day specimens, the impact on flexural strength was slightly less significant than at 28 days. During the initial hydration process, the load-bearing capacity of the cementitious composite is relatively weak, and the recycled PVC aggregate lacks the necessary compressive strength to support early load-bearing demands. However, as an elastomer, recycled PVC aggregate exhibits good tensile capacity, which contributes to the early flexural strength. This property partially offsets the negative impact on flexural strength compared to compressive strength.

3.6. Relative dynamic elastic modulus

The effect of recycled PVC aggregate on the dynamic elastic modulus of cementitious composite is shown in Fig. 14. With an increased substitution rate of recycled PVC aggregate, the relative dynamic elastic modulus of R-PVC-CC gradually decreased. When the replacement ratio reached 50 %, the relative dynamic elastic modulus decreased by 12.2 %.

The relative dynamic elastic modulus decreased by 12.2 % at 50 % PVC replacement. This reduction is primarily due to the lower stiffness of PVC compared to the cement matrix, which causes PVC particles to deform more easily under load. As demonstrated in the SEM analysis, the presence of microcracks and voids at the interface contributes to a less rigid structure, reducing the material's ability to recover after deformation. This finding aligns with the results of Ochi et al. [23], who reported similar reductions in modulus with plastic aggregate inclusion.

The observed trend of decreasing dynamic elastic modulus with higher recycled PVC content can be explained by the intrinsic material properties and the mixing process. The surface of recycled PVC aggregates tends to be irregular and less hydrophilic compared to traditional aggregates like quartz sand. This irregularity and hydrophobicity lead to poor bonding between the PVC particles and the cement matrix. As a result, more air is trapped during mixing, increasing the composite's overall porosity. Porosity is a critical factor affecting the mechanical properties of cementitious composites. Increased porosity reduces the load-bearing capacity and stiffness of the composite. The air pockets act as weak points within the material, disrupting the continuity and leading to higher deformability under load. The introduction of air and the consequent porosity also reduce the effectiveness of load transfer across the composite, contributing to the observed decrease in dynamic elastic modulus.

3.7. Resistance to dissolution and chloride ion penetration

The results of NH4Cl dissolution in R-PVC-CC are shown in Table 8. Fig. 15 presents the results of neutralization erosion, while Fig. 16 displays the outcomes of chloride ion erosion. With an increase in the recycled PVC aggregate admixture, the erosion depth gradually decreases, indicating an improvement in the resistance of R-PVC-CC to chloride ion erosion. Fig. 17 shows the erosion results of the specimens after 56 days of exposure.

The resistance to chloride ion penetration improved as the PVC content increased, with an 8.1 % reduction in penetration depth at 50 % replacement. This improvement is linked to the hydrophobic nature of PVC, which creates a less connected pore network, reducing chloride ingress. Similar trends were observed by Fakhri et al. [25] in asphalt mixtures with PVC. However, the increased porosity also affects mechanical properties, leading to trade-offs between durability and strength, as noted earlier.

The enhanced corrosion resistance of the cementitious composite with recycled PVC can be attributed to several key factors. Firstly, the hydrophobic nature of recycled PVC aggregates leads to the formation of tiny air bubbles on the surface of the aggregates during the mixing process. After hardening, the interface transition zone between the recycled PVC aggregates and the cement matrix contains many small pores. These pores disrupt the uniformity of capillaries within the cement matrix, hindering the diffusion of chloride ions and the neutralization process in the presence of NH4Cl. As a result, the addition of recycled PVC aggregates enhances the resistance of the cementitious composite to chloride ion penetration and improves durability.

Secondly, recycled PVC aggregates refine the internal pore structure of the cement matrix and alter the moisture transport pathways within the composite. Since water penetration is closely linked to the ingress of chloride ions, the altered pathways impede water transport, thereby reducing chloride ion penetration. This mechanism contributes to the observed decrease in neutralization depth and improved resistance to chloride ion attack in the cementitious composite with higher recycled PVC content.

3.8. Freeze-thaw resistance test

The mass loss rate of the R-PVC-CC after freeze-thaw cycles is shown in Fig. 18. The mass loss rate decreases with the increase of the recycled PVC aggregates. When the number of freeze-thaw cycles reached 100, the mass loss rates of the test groups with 5 %, 10 %, 20 %, 30 %, and 50 % recycled PVC aggregates decreased by 5.2 %, 4.6 %, 4.9 %, 4.3 % and 4.1 %, respectively. The mass loss rates of the recycled PVC groups were all lower than the 5 % of the control NM group. The frost resistance of the specimen was improved. There are several reasons for the following. First, during the cementitious composite mixing process, many tiny air bubbles are attached to the surface of the recycled PVC aggregate. The cementitious composite will lead to the existence of a large number of air pores at the contact interface between the recycled PVC and cement stone. These additional air pores can relieve the pressure inside the cement matrix due to thermal expansion and contraction during the cementitious composite's freezing and thawing process and relieve the capillary water's infiltration pressure. Second, during the freezing process, the water in the internal pores of the cementitious composite condenses into ice. When cementitious composite freezes, the water will generate a specific expansion stress, which makes the

Table 8
Erosion depth of R-PVC-CC.

Specimen number	Neutralization depth			Cl-erosion depth		
	28d	56d	90d	28d	56d	90d
NM	4.67	6.62	8.30	4.89	7.03	7.68
PM−20–5	4.64	6.58	8.11	4.61	6.78	7.35
PM−20–10	4.57	6.33	7.67	4.38	6.40	7.06
PM−20–20	4.46	6.22	7.20	4.21	6.21	6.61
PM−20–30	4.38	5.93	6.89	4.07	6.03	6.41
PM−20–50	4.30	5.69	6.53	3.81	5.67	6.01

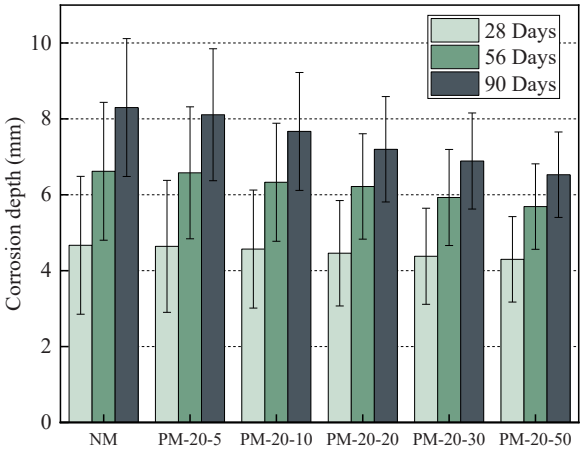


Fig. 15. Depth of the neutralization erosion.

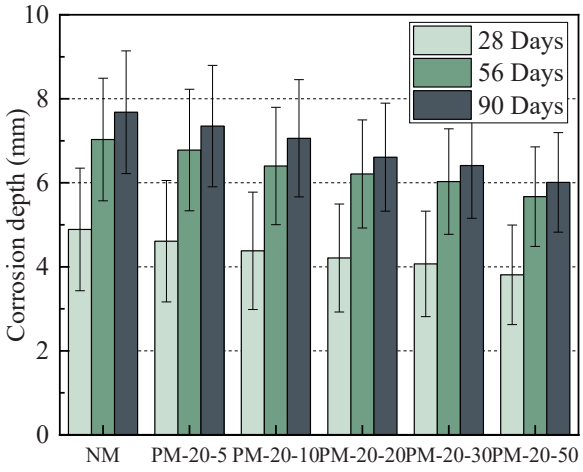


Fig. 16. Depth of the chloride ion erosion.

mortar volume larger. The recycled PVC aggregates have a certain elasticity. The volume contraction of recycled PVC aggregates in the process of mortar volume expansion can avoid stress concentration in the interface transition zone of cementitious composite and prevent the generation and development of cracks inside the cement matrix. Therefore, incorporating recycled PVC improves the freeze-thaw resistance of the cement matrix.

The transformation of the relative dynamic modulus of elasticity after freeze-thaw cycles is shown in Fig. 19. The relative dynamic modulus of elasticity of NM decreased by 10.4 % after 100 freeze-thaw cycles of cementitious composite. The relative dynamic modulus of elasticity of 8.5 %, 7.9 %, 6.7 %, 6.0 %, and 5.9 % for the test groups with 5 %, 10 %, 20 %, 30 %, and 50 % of recycled PVC aggregates replacement, respectively. Therefore, the admixture of recycled PVC aggregate alleviates the development of internal defects in the cementitious composite. The frost resistance of the cementitious composite is improved, and the best enhancement effect

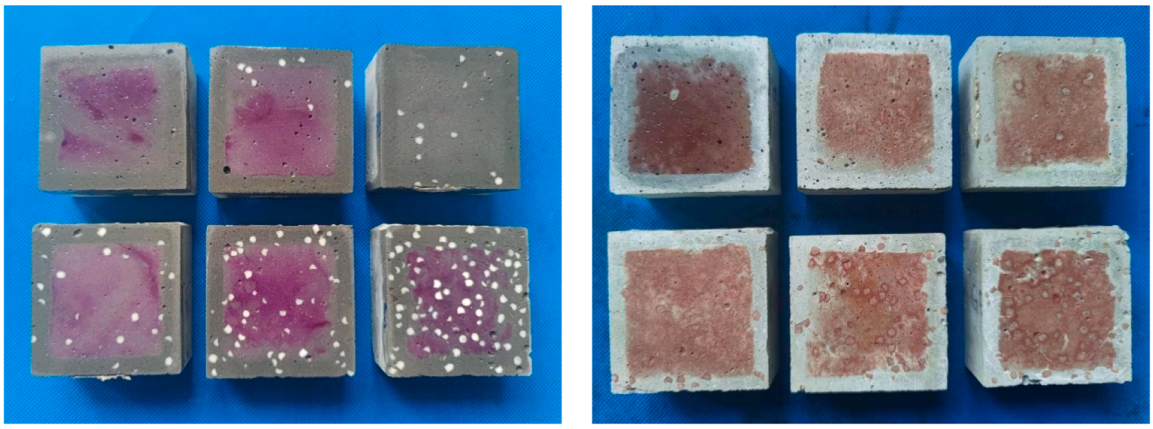


Fig. 17. Specimens after 56d of erosion: (a) Neutralization depth; (b) Cl-erosion depth.

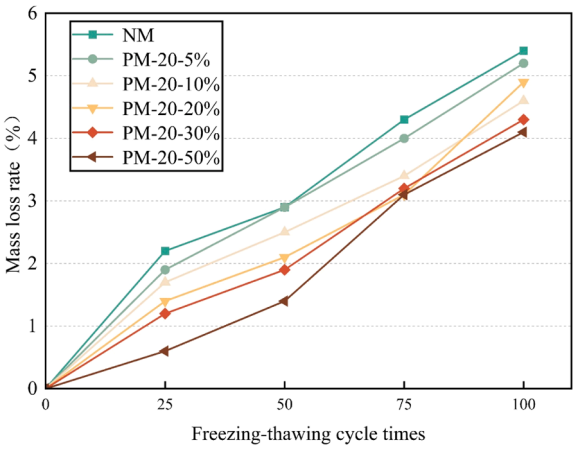


Fig. 18. Mass loss before and after freeze-thaw cycles.

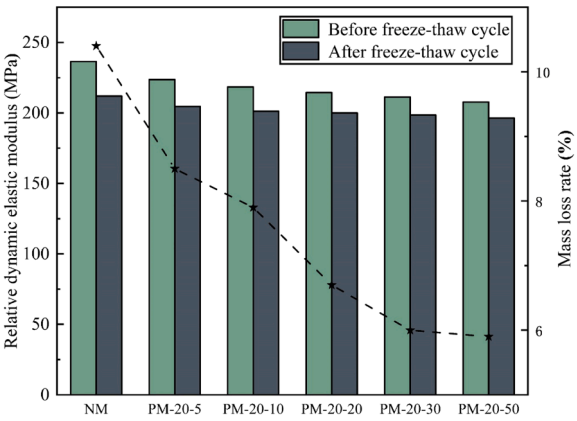


Fig. 19. Relative dynamic modulus of elasticity and its mass loss rate before and after freeze-thaw cycles.

is achieved when the recycled PVC aggregates are replaced at 50 %. In general, the mass loss rate and relative dynamic elastic modulus indexes show that recycled PVC aggregates can improve the freeze-thaw resistance of the cementitious composite more significantly. With the increase in the recycled PVC aggregate replacement rate, the freeze-thaw resistance of cementitious composite was significantly improved.

3.9. Scanning electron microscope

SEM images revealed increased porosity and microcracks at higher PVC replacement levels, particularly at the interface transition zone (ITZ). These microstructural observations explain the reduced mechanical properties, as the ITZ is the weakest area in the composite, and cracks initiate more easily at these points. Additionally, the increased porosity contributes to higher water absorption, as discussed earlier, which further compromises the composite's overall strength. The hydrophobic nature of PVC also plays a role in the formation of weak bonds at the ITZ, leading to a higher concentration of voids. This observation aligns with the findings of Reis et al. [24], where plastic aggregates resulted in similar microstructural issues in cementitious composites. Fig. 20 illustrates the microstructure of the recycled PVC-cement paste interface.

From Fig. 20, it is evident that the ITZ of the 20-mesh recycled PVC aggregate-cement paste is much less dense than that of the quartz sand-cement paste ITZ, with noticeable cracks present. Comparing the images in Fig. 20, the 20-mesh recycled PVC-cement paste ITZ shows the least densification and the widest cracks. Fig. 20(b) shows that the quartz sand-cement paste ITZ has the smoothest fracture surface, with only one main crack. The structure on both sides of the ITZ is dense, with fewer cracks, and the surrounding cementitious products are well-distributed, effectively filling cracks and voids outside the ITZ. In contrast, the recycled PVC-cement paste ITZ exhibits irregular cracks and numerous pores, indicating a lower bond strength compared to the quartz sand-cement paste ITZ. After the cement mortar is poured, the internal mortar undergoes hydration, releasing a significant amount of heat. This causes temperature differences within the cement matrix, leading to a decrease in internal moisture content. As the moisture evaporates, the internal mortar experiences drying shrinkage, resulting in cracks that affect the macro mechanical properties of the cement mortar. Fig. 21 shows the cracks in the recycled PVC cement matrix, with Fig. 21(a) illustrating the 20-mesh recycled PVC cement matrix cracks, and Fig. 21(b) showing the crack characteristics of the control group cement matrix.

Fig. 21(a) reveals that the 20-mesh recycled PVC, due to its material properties and larger particle size, lacks hydration products on its surface and does not bond tightly with the cement matrix, resulting in wider cracks in the ITZ. In Fig. 21(b), it is clear that the control group cement matrix has numerous hydration products from the cementitious materials accumulated around the internal cracks. The surrounding network of C-S-H gel, needle-like ettringite, and Ca(OH)2 crystals are tightly connected, providing some degree of blockage and coverage to the internal cracks.

When the cementitious composite is cast, the internal mortar will undergo a hydration reaction. The heat of hydration released by the hydration reaction causes a temperature difference between the interior and surface of the specimen, which results in a sharp decrease in the moisture content inside the cementitious composite. The moisture loss inside the specimen causes shrinkage and deformation of the cementitious composite. This situation can cause cracks in the specimens and affect the macroscopic mechanical properties of the cementitious composite. The control group accumulated many hydration products around the internal cracks. The net-like C-S-H gels, needle-like calcium alumina, and $Ca(OH)_2$ crystals around the cracks were more tightly connected, which played a role in plugging and covering the internal cracks. As shown in Fig. 21 the bond with the cement matrix is not tight. The interface transition zone produces cracks of significant width.

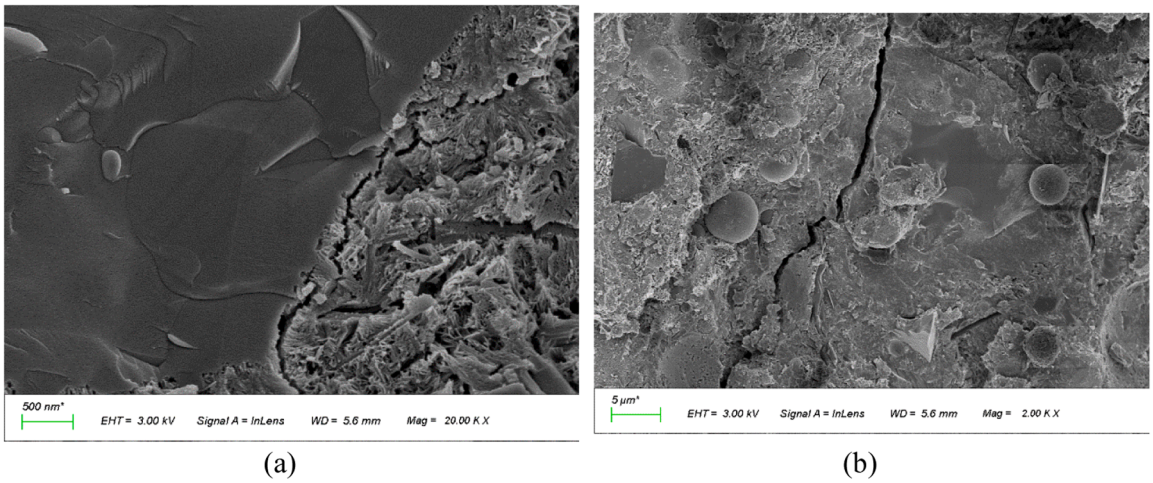


Fig. 20. Interface characteristics of fine aggregates and cement: (a) Interface between R-PVC and cement stone; (b) Interface between sand and cement stone.

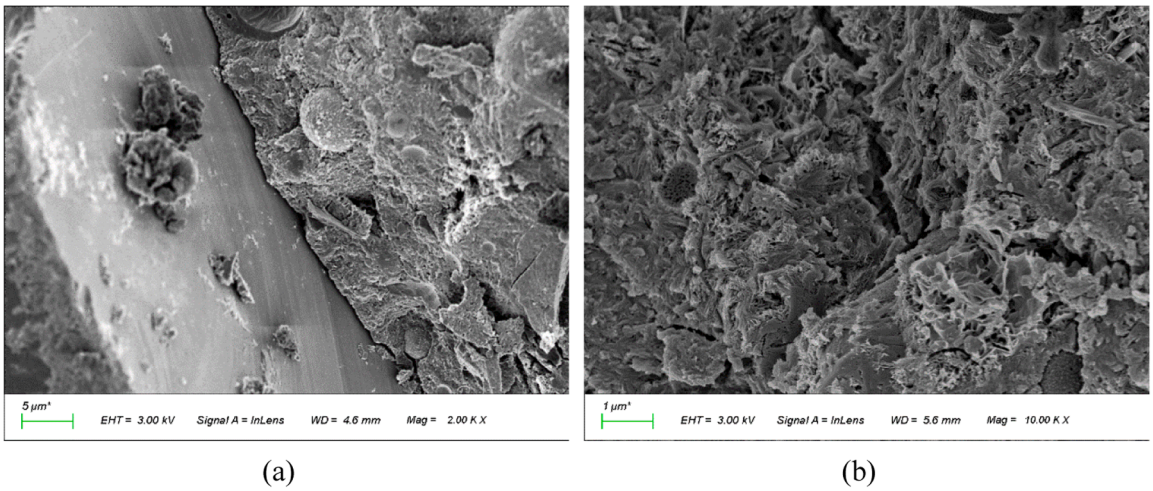


Fig. 21. Characteristics of cracks in cementitious composite.

4. Conclusion

This study innovatively explored the feasibility of incorporating recycled polyvinyl chloride (PVC) aggregates as substitutes for fine aggregates in cementitious composites, focusing on their mechanical properties, durability, and microstructural behavior. The key conclusions are as follows:

(1) The inclusion of recycled PVC aggregates significantly affected the mechanical properties of the cementitious composite. At 50 % PVC replacement, compressive strength decreased by 28.8 % and flexural strength dropped by 23.5 %. These reductions are primarily due to the poor bonding between PVC and the cement matrix, as observed in SEM analysis. The mechanical properties degrade at approximately 6.5 % for each additional 10 % PVC replacement in compressive strength, underscoring the need for careful balance between strength and sustainability.
(2) Despite the reduction in mechanical strength, the inclusion of PVC aggregates enhanced the composite's durability. Chloride ion penetration resistance improved by 8.1 % at 50 % PVC replacement, along with better freeze-thaw resistance. These improvements are attributed to the hydrophobic nature of PVC, which reduces capillary connectivity, thereby enhancing the composite's resistance to environmental degradation.
(3) SEM analysis revealed a weakened interface transition zone (ITZ) between the recycled PVC and the cement matrix, contributing to reduced mechanical strength. However, the increased porosity due to PVC inclusion also improved resistance to chloride ion ingress by disrupting capillary pathways, indicating a complex relationship between microstructural changes and overall performance.
(4) Further studies should explore methods to improve the bonding between PVC and the cement matrix, such as surface treatments or supplementary materials. Additionally, extended long-term durability tests, including freeze-thaw cycles and sulfate resistance, would provide further insights for practical applications.

CRediT authorship contribution statement

Limin Xu: Writing – review & editing, Resources. **Haoran Guo:** Resources, Investigation. **Zhaoxu Lv:** Writing – original draft, Formal analysis, Data curation, Conceptualization. **Caiqian Yang:** Writing – review & editing, Visualization, Validation, Supervision. **Yun Wan:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources. **Junshi Li:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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